Journal of Clinical Epidemiology 195 (2026) 112258

ORIGINAL RESEARCH

A multinomial hierarchical model for meta-analysis of diagnostic test accuracy of ordered or unordered multicategory tests

Nilotpal Chowdhury[a,*], Mohit Jadli[b,1], Meriyam Jahan[a,1], Mohan Kumar Verma[b,1]

[a]*Department of Pathology and Laboratory Medicine, All India Institute of Medical Sciences, Rishikesh, India*

[b]*Multidisciplinary Research Unit, All India Institute of Medical Sciences, Rishikesh, India*

Accepted 24 March 2026; Published online 2 April 2026

Abstract

Background and Objective: Multicategory diagnostic tests are frequently utilized in clinical practice. Estimation of the predictive value of each category is an essential part of these tests, yet they are often wrongly reported due to inappropriate analysis as a proportion ignoring pretest probability or prevalence. Also, some systems do not have fully ordered categories, making meta-analysis difficult. Analyses are often performed under the assumption that data are continuous, a premise that may not be appropriate and can result in unreliable confidence or credible intervals. These shortcomings can be rectified by multinomial mixed models, which can model both categorical as well as ordinal data, and estimation of predictive values through likelihood ratios (LRs). However, such multinomial models not requiring ordering have rarely, if at all, been used as a meta-analytical tool. This study aimed to employ a multinomial hierarchical linear mixed model appropriate for both ordered as well as partially ordered categories for such an analysis.

Methods: The proposed Bayesian multinomial mixed model estimates the proportions of each diagnostic category among cases with and without disease. This model was run on data extracted from a previous meta-analysis of the Milan System for Reporting Salivary Gland Cytopathology, which only partially follows ordered categories. The model was checked for proper mixing. From the posterior draws, the proportions of each category and the LRs were estimated. Fit of the data was checked by posterior predictive checks for location and width of distribution.

Results: The Monte Carlo Markov chains mixed well. The predictions of the model fitted the data well. From this, we could estimate the LR and predictive values, which were more informative than the previously reported Risk of Malignancy. Sensitivities and specificities could be calculated for different thresholds after ordering by the risk information derived from the LRs. Interpretation was aided by graphical examination of the predictive value to the prevalence of each category.

Conclusion: The proposed model is a relatively assumption free method that should be useful in meta-analysis of multicategory diagnostic tests if combined with subsequent LR estimation.

Keywords: Systematic review; Diagnostic test accuracy; Meta-analysis; Sensitivity and specificity; Predictive value of tests; Pretest probability

1. Introduction

Diagnostic tests employing multicategory diagnostic ratings to convey their results are common in diagnostic practice. These systems assign patients into multiple diagnostic levels or categories having distinct probabilities of disease. Examples of such systems include the Bethesda system for thyroid cytopathology [1] and the Yokohama system for breast cytopathology [2].

For meta-analysis of ordered multicategory rating systems, estimation of sensitivity and specificity by bivariate estimation separately at each threshold [3,4], or simultaneous joint estimation at multiple thresholds [5−7], has been described. From these estimates, we can estimate post-test probabilities, also known as predictive values (PVs), for the different thresholds, ie, $Pr(Disease|level \geq k)$ [8−11].

However, the PV of each individual level, ie, $Pr(Disease|level = k)$, is more useful than the PV above a

[1] Mohit Jadli, Meriyam Jahan, and Mohan Kumar Verma have made equal contribution to the work.

* Corresponding author. Department of Pathology and Laboratory Medicine, All India Institute of Medical Sciences, Rishikesh 249203, India.

E-mail address: nilotpal.chowdhury@gmail.com (N. Chowdhury).

https://doi.org/10.1016/j.jclinepi.2026.112258

What is new?

Key findings

- A Bayesian model for meta-analysis of a multicategory diagnostic test is proposed which is applicable to both ordered and unordered categories.

What this adds to what is known

- Meta-analysis of diagnostic accuracy of multicategory systems are often not possible by current methods due to failure to meet assumptions.
- This model is applicable to any multicategory test, whether current methods fail or succeed.
- Predictive values are obtained from the category-specific LR, which are preferable to commonly reported pooled predictive values.

What is the implication and what should change now?

- This method should enable valid meta-analysis of most multicategory diagnostic assessment tests.

threshold. Since a clinician often does not have the reference test at the time of decision making, it is these PV that serves as a guidance for decisions. These PV should be estimated from the level specific Likelihood ratios (LRs), [12] and is also explained in the methods. In the more common dichotomous test setting, these level specific PV become positive and negative PV (PPV and NPV) for the positive and negative index tests. A valid meta-analysis should include estimation of sensitivity, specificity, or category probabilities followed by LR and PV. Such analysis for multicategory tests is rare.

Many meta-analytical methods to assess multiple thresholds assume an underlying continuous variable with a defined statistical distribution, commonly Gaussian [5], logistic [5,13], transformed logistic [14], or Weibull [13]. There is poor statistical coverage when these distributional assumptions are not met [15], which is often the case for multinomial or ordered categories. Also, all known methods become invalid if there is uncertainty regarding ordering of the categories. Examples of systems with partially unordered categories include the Milan System for Reporting Salivary Gland Cytopathology (Table 1) [16] or the World Health Organization system of reporting pancreatico-biliary cytopathology [17]. A multinomial model would be valid in all these conditions.

A pervasive reason for invalid meta-analysis is that many studies have used flawed methods for PV, disregarding its dependence on prevalence. All PV follow the same relation with prevalence as PPV and NPV. Many studies analyze PV of multicategory tests as simple proportions without accounting for prevalence or pretest probability [9,18]. Others studies perform naïve pooling of the PV or just report the range of PV reported in constituent studies [10,11,19,20]. All these problems may be addressed using a multinomial mixed model with subsequent estimation of category specific LR and PV [12].

1.1. Motivating example

Fine needle aspiration biopsy (FNAB) is commonly used to diagnose salivary gland lesions. The International Academy of Cytology has proposed the seven-tier Milan System for Reporting of Salivary Gland Cytopathology (Table 1) [16,21]. The Milan system primarily aims to stratify patients with salivary gland lesions into distinct risk categories for cancer by Risk of Malignancy (ROM). ROMs are PVs. Also, the benign neoplastic lesions differ in treatment given, with benign neoplastic lesions have a different treatment than non-neoplastic lesions. Thus, nonmalignant neoplastic lesions have been assigned a distinct group. The risk hierarchy for malignancy does not always correspond to the grouping applied. There are two benign groups (non-neoplastic and neoplasm-benign) assigned to different categories. Atypia of undetermined significance (AUS) and salivary gland neoplasm of uncertain malignant potential (SUMP), though different categories, have close suggested ROM. The nondiagnostic category represents failed tests and should be excluded for analysis of accuracy. There is uncertainty in ordering the categories by cancer risk. This uncertainty possibly resulted in meta-analyses of the Milan System removing AUS and SUMP from their analysis of sensitivity and specificity [10,22]. We attempted to reanalyze a published meta-analysis [22] for the accuracy of the Milan system in diagnosing salivary gland cancer. Histopathology served as the reference test. To bypass difficulties in ordering categories, we propose a Bayesian multinomial mixed effects model. From the posterior samples, we estimate the category-specific LR and PV, which are clinically useful measures for clinicians.

2. Materials and methods

2.1. The Bayesian hierarchical multinomial model

We introduce a Bayesian multinomial generalization the bivariate generalized linear mixed effects model (GLMM) of Chu and Cole [3] currently used for diagnostic tests with a single threshold. Like the Chu and Cole model, we consider n studies, each divided into two strata for diseased (or positive) and nondiseased (or negative) populations. Instead of modeling binomial outcomes (index test positive vs negative), we model the response as multinomial counts, with each count belonging to one of several categories. Using

Table 1. The Milan System for reporting salivary gland cytopathology (second edition)

Category	Estimated risk of malignancy	Remarks
I. Nondiagnostic	15% (initially estimated at 25% in 1st edition)	Represents failed test by its definition. Should not have a predictive value
II. Non-neoplastic	11% (revised from 10% in 1st edition)	May have a lower risk than reported since many negative cases are not followed up surgically.
III. Atypia of Undetermined Significance (AUS)	30% (revised from 20% in 1st edition)	Estimates on limited data
IV. Neoplastic		
a. Benign	<5%	
b. SUMP	35%	Estimates on limited data
V. Suspicious for malignancy	83% (revised from 60% in 1st edition)	
VI. Malignant	>98% (revised from >90% in 1st edition)	

SUMP, salivary gland neoplasm of uncertain malignant potential.

multinomial logistic regression, we model the log-odds of each category relative to a reference category (baseline category). These log-odds are predicted by disease status within each stratum.

The fixed-effects part of the model consists of the regression coefficients describing the effect of disease status on the log-odds of each nonreference category. For each study, the study-specific deviations (random effects) from these fixed effects are assumed to follow a multivariate normal distribution.

The log-odds for diseased and nondiseased strata may be correlated within each study. The random effects allow the multinomial log-odds (one for each nonreference category) to vary across studies, capturing between-study heterogeneity. From these study-level and population-level parameters, we obtain estimates of the probabilities of each outcome category in the diseased and nondiseased strata. The mathematical formulation of the model is given in Supplementary appendix 1.

LR is calculated as the ratio of the probability of assigned to category k with disease, to the probability of being assigned to category k without disease, ie, $LR_k = \frac{P(y=k)_{dis}}{P(y=k)_{ndis}}$.

The post-test probability, ie, the PV, for an individual category is then determined by $Pr(Diseased|level = k) = \frac{1}{1+\frac{1-p}{p.LR_k}}$, where p is the pretest probability or prevalence. A reasonable range of prevalences are chosen based on prior knowledge. For any specified prevalence, credible intervals (CIs) for the PV can be derived by just substituting the upper and lower credible limits of the LR into the above equation [23].

2.2. Data and methodology

Data for the current reanalysis were extracted from the original study (Supplementary Table 1). The original study aimed to assess primarily prospective studies; retrospective studies were also assessed for comparison. The index test was FNAB, which was reported before the reference test. Histopathology was the reference standard, with a histopathology diagnosis of cancer considered positive and noncancer considered negative. We conducted the reanalysis by examining the prospective and retrospective studies separately and combined. Neoplastic-benign and SUMP groups were treated as separate categories. To estimate the distribution of multinomial categories, Bayesian multivariate modeling was employed using the R packages "brms" [24] and "cmdstanr" [25]. Four Monte Carlo Markov Chains (MCMC) chains of 6000 iterations each were run, of which the initial 2000 iterations were designated for warmup. The priors used were normal prior with a mean of 0 and SD of 5 for regression coefficients, an Lewandowski-Kurowicka-Joe prior with shape parameter 1 for correlation between random effects and half-Student t distribution with 3 degrees of freedom for SDs of random effects.

Convergence was assessed visually using trace plots and using standard diagnostics (eg $\widehat{R} \cong 1$). The LR for each category was calculated, and the 95% CIs of LRs estimated from the posterior draws. Subsequently, the PV of each category was calculated for prevalences of 5%, 15%, 30%, and 50%, and also graphically examined for the full range of prevalences from 0% to 100%. The categories were then ordered based on their LR, and sensitivities and specificities at each threshold estimated along with the area under the receiver operating characteristic curve.

The model fit was assessed on the predicted posterior distribution from the Bayesian model. The predicted probability of each category in each study was examined by boxplots and assessed whether they were compatible with the actual observed values. The variance of the predicted expected probability in each category across studies were compared to the corresponding observed variance of the category probabilities to assess for overdispersion. In addition, the spread of the predicted values was compared to the spread expected from the observed counts in each category and study as a rough check for overdispersion. The median of the predicted

Table 2. Likelihood ratios of the three meta-analyses performed

	Non-neoplastic	AUS	Neoplastic-benign	SUMP	Suspicious	Malignant
Prospective studies						
Estimate [95% CI]	0.40 [0.13, 1.67]	2.76 [0.28, 66.59]	0.06 [0.01, 0.30]	1.65 [0.39, 12.96]	44.79 [6.55, 601.1]	339.7 [45.07, 7132]
Prediction interval	[0.05, 133.5]	[0.05, 36.683]	[0.00, 99.21]	[0.11, 2228]	[0.93, 70,878]	[6.31, 765,391]
Retrospective studies						
Estimate [95% CI]	0.31 [0.13, 0.69]	1.62 [0.25, 7.39]	0.06 [0.02, 0.14]	1.30 [0.47, 3.48]	4.97 [1.42, 20.21]	101.2 [23.14, 2178]
Prediction interval	[0.05, 1.46]	[0.02, 45.00]	[0.00, 0.56]	[0.10, 12.56]	[0.20, 140.8]	[1.51, 31,073]
All studies						
Estimate [95% CI]	0.33 [0.17, 0.65]	2.25 [0.73, 9.42]	0.07 [0.03, 0.12]	1.51 [0.71, 3.62]	11.49 [4.08, 46.53]	151.8 [48.80, 1241]
Prediction interval	[0.08, 3.63]	[0.13, 229.2]	[0.01, 1.15]	[0.25, 41.36]	[0.61, 756.0]	[7.43, 18,795]

95% prediction interval is also given.
AUS, atypia of undetermined significance; SUMP, salivary gland neoplasm of uncertain malignant potential; CI, credible interval.

values was compared with the observed cell counts in each category having zero observations to assess for zero inflation.

3. Results

A summary of the results of the fit of the Milan System is given in Supplementary Table 2. The trace plots are given in Supplementary Figures1, S2 and S3 for the three different analyses. All estimated effects had R-hat values of 1.00, a large effective sample size, and satisfactory trace plots, suggesting effective mixing of chains. Significant heterogeneity between studies was suggested by many of the random effect showing large SDs. Heterogeneity between the prospective and retrospective studies were also observed on posterior draws (Supplementary Figure 4). Prospective and retrospective studies were then combined for a third analysis.

LR for all categories were calculated from posterior draws (Table 2). The categories were ordered according to their risk as deduced from the LRs. Such ordering created valid thresholds between adjacent ordered categories. Sensitivity and specificity for all such thresholds were calculated (Table 3). Ninety-five percent prediction intervals (PIs) were also estimated.

PVs at prevalences commonly encountered by clinicians (5%, 15%, 30%, and 50%) are given in Table 4 and supplementary Table 3 along with the ROM reported in the original study [22]. For a more general picture of PV, a plot of the PV vs prevalence is given in Figure 1.

Posterior predictive checks including predicted vs observed category probabilities and within- and between-category variances and zero counts, were satisfactory, implying good fit (Supplementary Figures 5—10).

4. Discussion

The results of the LRs show neoplastic-benign, non-neoplastic, suspicious, and malignant categories were

Table 3. Sensitivities and specificities of thresholds obtained after reordering of the categories according to the Likelihood ratios

	AUS + SUMP + suspicious + malignant considered positive		Suspicious + Malignant considered positive		Only malignant considered positive		
	Sensitivity	Specificity	Sensitivity	Specificity	Sensitivity	Specificity	AUC
Prospective studies							
Estimate [95% CI]	0.90 [0.81, 0.96]	0.83 [0.35, 0.97]	0.63 [0.43, 0.78]	1 [0.98, 1]	0.50 [0.30, 0.67]	1 [0.99, 1]	0.93 [0.79, 0.98]
Prediction interval	[0.46, 0.99]	[0, 1]	[0.10, 0.91]	[0.87, 1]	[0.04, 0.84]	[0.96, 1]	[0.46, 1]
Retrospective studies							
Estimate [95% CI]	0.91 [0.85, 0.96]	0.88 [0.80, 0.93]	0.80 [0.69, 0.88]	0.96 [0.90, 0.99]	0.62 [0.46, 0.75]	0.99 [0.97, 1]	0.95 [0.90, 0.97]
Prediction interval	[0.68, 0.98]	[0.31, 0.96]	[0.26, 0.94]	[0.44, 1]	[0.11, 0.84]	[0.73, 1]	[0.64, 0.98]
All studies							
Estimate [95% CI]	0.90 [0.86, 0.94]	0.87 [0.75, 0.94]	0.72 [0.63, 0.80]	0.98 [0.96, 0.99]	0.57 [0.47, 0.67]	1 [0.99, 1]	0.94 [0.91, 0.97]
Prediction interval	[0.76, 0.97]	[0.1, 0.99]	[0.31, 0.9]	[0.64, 1]	[0.17, 0.80]	[0.93, 1]	[0.65, 0.98]

95% prediction interval is also given.
CI, credible interval; AUC, area under the ROC curve; AUS, atypia of undetermined significance; SUMP, salivary gland neoplasm of uncertain malignant potential.

Table 4. Predictive values for each category at prevalences of 0.05, 0.15, 0.3, and 0.5 with 95% credible intervals

p	Non-neoplastic	AUS	Neoplastic-benign	SUMP	Suspicious	Malignant
Prospective studies						
0.05	0.02 [0.01, 0.08]	0.13 [0.01, 0.78]	0.00 [0.00, 0.02]	0.08 [0.02, 0.41]	0.70 [0.26, 0.97]	0.95 [0.7, 1]
0.15	0.07 [0.02, 0.23]	0.33 [0.05, 0.92]	0.01 [0.00, 0.05]	0.23 [0.07, 0.7]	0.89 [0.54, 0.99]	0.98 [0.89, 1]
0.3	0.15 [0.05, 0.42]	0.54 [0.11, 0.97]	0.03 [0.00, 0.11]	0.41 [0.14, 0.85]	0.95 [0.74, 1]	0.99 [0.95, 1]
0.5	0.29 [0.12, 0.62]	0.73 [0.22, 0.99]	0.06 [0.01, 0.23]	0.62 [0.28, 0.93]	0.98 [0.87, 1]	1 [0.98, 1]
Retrospective studies						
0.05	0.02 [0.01, 0.03]	0.08 [0.01, 0.28]	0.00 [0.00, 0.01]	0.06 [0.02, 0.15]	0.21 [0.07, 0.52]	0.84 [0.55, 0.99]
0.15	0.05 [0.02, 0.11]	0.22 [0.04, 0.57]	0.01 [0.00, 0.02]	0.19 [0.08, 0.38]	0.47 [0.2, 0.78]	0.95 [0.80, 1]
0.3	0.12 [0.05, 0.23]	0.41 [0.10, 0.76]	0.03 [0.01, 0.06]	0.36 [0.17, 0.6]	0.68 [0.38, 0.9]	0.98 [0.91, 1]
0.5	0.24 [0.11, 0.41]	0.62 [0.2, 0.88]	0.06 [0.02, 0.12]	0.57 [0.32, 0.78]	0.83 [0.59, 0.95]	0.99 [0.96, 1]
All studies						
0.05	0.02 [0.01, 0.03]	0.11 [0.04, 0.33]	0.00 [0.00, 0.01]	0.07 [0.04, 0.16]	0.38 [0.18, 0.71]	0.89 [0.72, 0.98]
0.15	0.05 [0.03, 0.10]	0.28 [0.11, 0.62]	0.01 [0.00, 0.02]	0.21 [0.11, 0.39]	0.67 [0.42, 0.89]	0.96 [0.90, 1]
0.3	0.12 [0.07, 0.22]	0.49 [0.24, 0.8]	0.03 [0.01, 0.05]	0.39 [0.23, 0.61]	0.83 [0.64, 0.95]	0.98 [0.95, 1]
0.5	0.25 [0.14, 0.39]	0.69 [0.42, 0.9]	0.06 [0.03, 0.11]	0.60 [0.42, 0.78]	0.92 [0.8, 0.98]	0.99 [0.98, 1]
Values obtained by simple pooling						
Pro	0.09 [0.02, 0.17]	0.35 [0.18, 0.52]	0.02 [0.01, 0.04]	0.37 [0.28, 0.45]	0.86 [0.74, 0.98]	0.97 [0.94, 1]
Retro	0.08 [0.02, 0.14]	0.40 [0.21, 0.58]	0.02 [0.01, 0.04]	0.31 [0.18, 0.45]	0.66 [0.50, 0.82]	0.97 [0.94, 1]

These values differ markedly at low and high prevalence from those calculated by a simple meta-analysis of predictive values as proportions without accounting for the prevalence.

AUS, atypia of undetermined significance; SUMP, aalivary gland neoplasm of uncertain malignant potential; Pro, prospective; Retro, retrospective.

associated with distinct increasing risk profiles. AUS and SUMP categories are largely overlapping. The final order obtained was: neoplastic-benign < non-neoplastic < SUMP/AUS < suspicious < malignant. This allows estimation of threshold specific sensitivities, specificities, and AUC.

Table 4 presenting the actual PV for malignancy across a range of prevalences, indicates that the reported ROM suggested by the authors of the Milan system apply primarily to cases with a 15%-30% prevalence. However, the probability that a salivary gland lesion is malignant varies significantly by anatomical site—approximately 20%-25% for parotid glands, 35%-45% for submandibular glands, and over 50% for other salivary glands [26]—as well as by clinical setting and prebiopsy laboratory and imaging studies. As prevalence increases, so does the PV. The varying prevalences across anatomical sites make a single estimate of ROM uninformative and even misleading. Updating the pretest probabilities for patients depending on the anatomical site and other clinical information and subsequently calculating PV, should give more valid risk estimates for malignancy.

Assessment of PV across the full range of clinical scenarios or prevalences, rather than on few discrete data points, is aided by the plot of PV against prevalence (Fig 1) [27]. Each curve on the plot displays the median predicted PV, bound by a CI that visualizes the uncertainty around that estimate. A vertical movement from a known pretest probability or prevalence (x-axis) to the curve yields the updated PV (y-axis). A curve moving sharply toward the top-left corner signifies a result highly effective at ruling in the disease (strong positive association). A curve moving toward the bottom-right corner signifies a result highly effective at ruling out the disease (strong negative association). The width of the CI region is the visual marker of uncertainty; wider regions indicate lower confidence in the diagnostic power of that category.

If a confidence region crosses the central diagonal line (the line of no information, where post-test = pretest probability), the test result is considered nonsignificant. Distinct diagnostic categories should be well separated; overlapping confidence regions imply that the risk profiles of the categories are not significantly different statistically. The confidence regions for benign atypical, suspicious, and malignant categories show clear separation when the full dataset is analyzed, suggesting that they are useful, distinct categories. The AUS category fully overlaps with SUMP, occupying practically the same predictive space. This suggests they provide indistinguishable clinical information and should be considered for combination.

The random effects show the between-study heterogeneity. Assessment of magnitude of heterogeneity is helped by comparison against the SD of the latent distribution of fixed effects (which is $\frac{\pi}{\sqrt{3}} \cong 1.81$ due to the logit link). Many

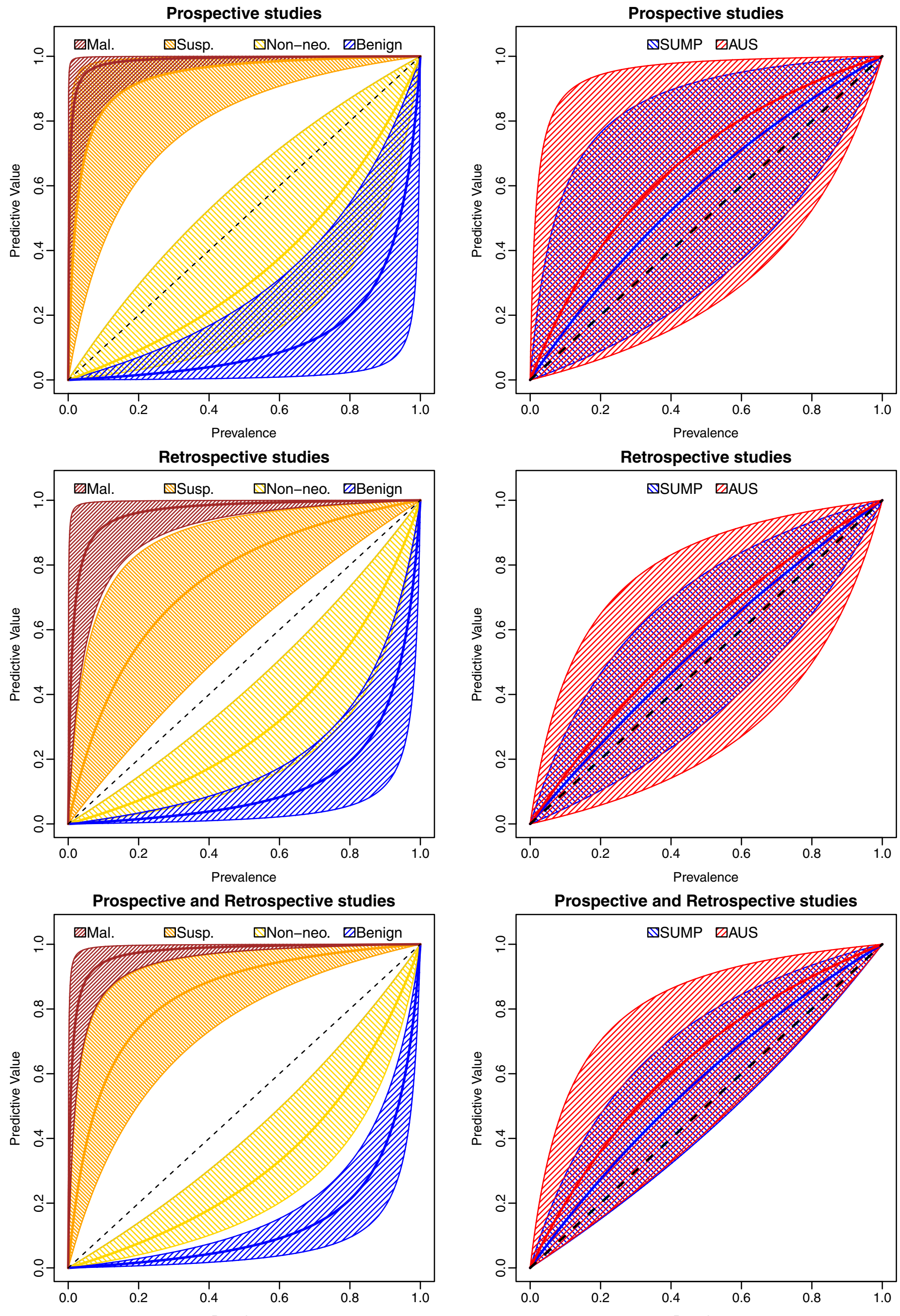

Figure 1. Showing the predictive values vs prevalence for the neoplastic-benign, non-neoplastic, suspicious, and malignant categories on the left panel, and the AUS and SUMP categories on the right panel. All three analyses (on prospective studies only, retrospective studies only, and combined analysis) have been included. For individual patients undergoing clinical testing, updated pretest probabilities may be used instead of the prevalence. AUS, atypia of undetermined significance; SUMP, salivary gland neoplasm of uncertain malignant potential; Mal., malignant; Susp.; suspicious; Non-neo., non-neoplastic.

random effects for the negative population were close to or exceeded this value, suggesting high heterogeneity. Subgroup comparison is possible by comparing the posterior distributions of group specific analysis (Table 2, Supplementary Table 2, and Supplementary Figure 4). Doing this, prospective studies showed a higher LR for most categories compared to the retrospective studies. This may be partially explained by the increased familiarity with the Milan System in prospective studies, which were done later than retrospective studies. High heterogeneity results in wide PIs of estimates.

These results are valid under the assumption that the sensitivities and specificities are independent of prevalence. This assumption may be tested by studies specifically addressing diagnostic accuracy in relatively narrow pretest probabilities, eg, in patients stratified by radiological assessment of cancer risk.

To the best of our knowledge, this is the first study to use a multinomial model suitable for meta-analysis of diagnostic tests comprising of unordered, partially ordered and ordered categories. For ordered categories, the Stoye model is an alternative [28]. The bivariate GLMM [3] can also be applied at each threshold to estimate threshold-specific sensitivities and specificities but is limited by absence of a valid method to calculate CI of LRs. The continuous response models of Steinhauser [5,29], Jones [7,14], and Hoyer [13,15,30] models may be used if ordered categorical data can be justifiably treated as continuous. All these models were unsuited for the initial part of the present meta-analysis since we were unsure of the correct category order.

Comparison between the proposed Bayesian multinomial model and the above-mentioned models was possible after ordering the categories. Two separate ordering criteria were used: first, according to the ROM suggested by the authors of the Milan System, and second, according to the LR suggested by the multinomial analysis (Supplementary Table 4). There was sharp discordance in the results between the multinomial model and the continuous response models. In both cases, the continuous models showed significant difference in LR between the AUS and SUMP categories, with their relative ordering depending on the predefined order chosen. The Stoye models showed largely overlapping confidence intervals for AUS and SUMP like the multinomial model, but the ordering followed the predefined scheme. The bivariate GLMM results ordered AUS and SUMP opposite to continuous or Stoye models.

Thus, the Stoye models were the only frequentist models that showed the practical equivalence of AUS and SUMP categories. They provide useful statistical inference if correct CIs are estimated, though the mean estimates are sensitive to the order assumed. The underlying Stoye models assume categorical response, even though categories are ordered for data preparation, which possibly explains their valid CIs.

The Steinhauser, Jones, and Hoyer models designed for continuous responses showed marked difference with the multinomial model [5,7,13−15,30]. The distribution of the predicted LR from these models were visualized graphically (Supplementary Figure 11) and contrasted with results derived from bivariate GLMM at each threshold. The continuous response models required a continuously decreasing LR in relation to false positive rate. If two adjacent categories have similar LR, such decreasing relation would be absent. In such a case, the predicted LR by continuous models will be pulled apart in opposite directions from their actual values, resulting in inaccurate inference. This may result in false reporting of significant difference between similar risk categories, as in the present case. Other model-specific distributional assumptions (eg normal, logistic etc.) are an additional limiting factor. Coverage is markedly reduced if these assumptions are not met [15].

Two models could not be included in the comparison above. The model by Hamza et al. [31] assumes linear relationship between the logits of the sensitivity and false-positive fraction. Successful optimization and results are dependent on initial starting values. For the present analysis, this method failed optimization even after multiple attempts at providing starting values. Putter [32] used a Poisson correlated gamma frailty model but did not provide a method for estimating LRs. Joint estimation of CI is complicated and may need nonparametric bootstrap, which may perform poorly for small number of studies.

The current software implementations of all the methods above also do not estimate the LR with CI. This requires separate extraction of source code followed by coding the delta method or parametric bootstrap. Delta method may yield inaccurate CIs at small sample sizes or extreme LR. The parametric bootstrap, especially for the Stoye models, are extremely time-consuming.

In contrast to the models outlined above, multinomial models do not even assume ordered categories. The categories just need to follow a multinomial distribution. Random effects need to follow normal distribution. For R users, the syntax is similar to "lme4" package currently used for bivariate analysis. Bayesian models use MCMC draws for CIs for LR, which avoids delta method approximations used by frequentist models [33]. Disadvantages include the longer time for analysis, possible delayed mixing of chains, and investment of time in selecting a proper prior and checking the posterior distribution.

In case of failure of the model to converge, bivariate meta-analysis treating each individual category as the positive category and all other categories as the negative category may yield valid LRs. Other methods like univariate pooling of LR have been proposed but are discouraged due to the possibility of obtaining impossible values [33,34].

This study is aimed to illustrate use of a meta-analytical model and not for new evidence generation. The demonstration of the ability of a Bayesian model to recover the parameter values needs a simulation or a simulation based calibration study [35], which would reveal the validity of the model as well as the limits and robustness of inference that can be drawn. The primary study focused on prospective studies; for evidence assessment, only the section on prospective studies is valid.

5. Conclusion

In conclusion, we have described a Bayesian hierarchical model suitable for meta-analysis of diagnostic tests with multiple ordered or unordered categories, allowing estimation of category-specific LRs and PVs. This model should be useful since it makes minimal assumptions about data, including ordinality, and leads to sound, interpretable results.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT and Google Gemini 3 in order to debug and help writing the R code, and Microsoft Copilot in order to edit the manuscript after preparation. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Nilotpal Chowdhury: Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Mohit Jadli:** Writing – review & editing, Visualization, Validation, Investigation, Data curation. **Meriyam Jahan:** Writing – review & editing, Visualization, Validation, Investigation, Data curation. **Mohan Kumar Verma:** Writing – review & editing, Visualization, Validation, Investigation, Data curation.

Declaration of competing interest

The authors declare no conflicts of interest.

Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2026.112258.

Data availability

The data are publicly available and uploaded as supplementary file.

References

[1] Cibas ES, Ali SZ. The 2017 bethesda System for reporting thyroid cytopathology. Thyroid 2017;27(11):1341–6.
[2] Field AS. Breast FNA biopsy cytology: current problems and the International Academy of Cytology Yokohama standardized reporting system. Cancer Cytopathol 2017;125(4):229–30.
[3] Chu H, Cole SR. Bivariate meta-analysis of sensitivity and specificity with sparse data: a generalized linear mixed model approach. J Clin Epidemiol 2006;59:1331–2.
[4] Takwoingi Y, Dendukuri N, Schiller I, Rücker G, Jones H, Partlett C, et al. Chapter 10: undertaking meta-analysis. In: Deeks J, Bossuyt P, Leeflang M, Takwoingi Y, editors. Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy Version 2.0 (updated July 2023). Cochrane; 2023. Available at: https://training.cochrane.org/handbook-diagnostic-test-accuracy/current. Accessed March 27, 2026.
[5] Steinhauser S, Schumacher M, Rücker G. Modelling multiple thresholds in meta-analysis of diagnostic test accuracy studies. BMC Med Res Methodol 2016;16:97.
[6] Macaskill P, Takwoingi Y, Deeks JJ, Gatsonis C. Chapter 9: understanding meta-analysis. In: Deeks JJ, Bossuyt PM, Leeflang M, Takwoingi Y, editors. Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy. Version 2.0 (updated July 2023). Cochrane; 2023. Available at: https://training.cochrane.org/handbook-diagnostic-test-accuracy/current. Accessed March 27, 2026.
[7] Zapf A, Albert C, Frömke C, Haase M, Hoyer A, Jones HE, et al. Meta-analysis of diagnostic accuracy studies with multiple thresholds: comparison of different approaches. Biometrical J 2021;63(4):699–711.
[8] Paul P, Azad S, Agrawal S, Rao S, Chowdhury N. Systematic review and meta-analysis of the diagnostic accuracy of the International Academy of Cytology Yokohama System for reporting breast fine-needle aspiration biopsy in diagnosing breast cancer. Acta Cytol 2023;67:1–16.
[9] Castellana M, Grani G, Radzina M, Guerra V, Giovanella L, Deandrea M, et al. Performance of EU-TIRADS in malignancy risk stratification of thyroid nodules: a meta-analysis. Eur J Endocrinol 2020;183(3):255–64.
[10] Wang Z, Zhao H, Guo H, An C. Application of the Milan System for reporting Salivary Gland Cytopathology: a systematic review and meta-analysis. Cancer Cytopathol 2022;130(11):849–59.
[11] Ahuja S, Fattahi-Darghlou M, Zaheer S, Ahuja R. Diagnostic accuracy of Papanicolaou Society of Cytopathology System for reporting respiratory cytology: a systematic review and meta-analysis. Acta Cytol 2024;68:450–64.
[12] Newman TB, Kohn MA, Evidence-Based Diagnosis. Evidence-Based Diagnosis An Introduction to Clinical Epidemiology. 2nd ed. Cambridge, NY: Cambridge University Press; 2020:47–74.
[13] Hoyer A, Kuss O. Meta-analysis of full ROC curves with flexible parametric distributions of diagnostic test values. Res Synth Methods 2020;11(2):301–13.
[14] Jones HE, Gatsonsis CA, Trikalinos TA, Welton NJ, Ades AE. Quantifying how diagnostic test accuracy depends on threshold in a meta-analysis. Stat Med 2019;38:4789.
[15] Zapf A, Frömke C, Hardt J, Rücker G, Voeltz D, Hoyer A. Meta-Analysis of diagnostic accuracy studies with multiple thresholds: comparison of approaches in a simulation Study. Biometrical J 2024;66(7):e202300101.
[16] Pusztaszeri M, Rossi ED, Baloch ZW, Faquin WC. Salivary gland fine needle aspiration and introduction of the Milan reporting System. Adv Anat Pathol 2019;26:84–92.
[17] Hoda RS, Arpin RN, Rosenbaum MW, Pitman MB. Risk of malignancy associated with diagnostic categories of the proposed World Health Organization International System for Reporting Pancreaticobiliary Cytopathology. Cancer Cytopathol 2022;130(3):195–201.
[18] Nikas IP, Vey JA, Proctor T, AlRawashdeh MM, Ishak A, Ko HM, et al. The use of the International Academy of Cytology Yokohama System for reporting breast fine-needle aspiration BiopsyA systematic review and meta-analysis. Am J Clin Pathol 2023;159:138–45.
[19] Vuong HG, Chung DGB, Ngo LM, Bui TQ, Hassell L, Jung CK, et al. The use of the bethesda System for reporting thyroid cytopathology in pediatric thyroid nodules: a meta-analysis. Thyroid 2021;31(8):1203–11.
[20] Hoda RS, Brachtel EF. International Academy of cytology Yokohama System for reporting breast fine-needle aspiration Biopsy Cytopathology: a review of predictive values and risks of malignancy. Acta Cytol 2019;63:292–301.
[21] Baloch Z, Lubin D, Katabi N, Wenig BM, Wojcik EM. The Milan System for reporting salivary Gland cytopathology. In: Faquin WC, Rossi ED, Baloch Z, Barkan GA, Foschini MP, Kurtycz DFI, et al,

editors. The Milan System for Reporting Salivary Gland Cytopathology, Second Edition. , 2nd ed. Cham: Springer Nature Switzerland AG; 2023:1−13.

[22] Lagerstam H, Kalfert D, Maleki Z, Kholová I. How the Milan System for Reporting Salivary Gland Cytopathology works in cytopathology practice: Meta-analysis of prospective studies and comparison with retrospective studies. Cancer Cytopathol 2024;132(7):447−57.

[23] Li J, Fine JP, Safdar N. Prevalence-dependent diagnostic accuracy measures. Stat Med 2007;26:3258−73.

[24] Bürkner P-C. Advanced bayesian multilevel modeling with the R package brms. R J 2018;10(1):395−411.

[25] Gabry J, Češnovar R, Johnson A, Bronder S. Cmdstanr: r interface to "CmdStan." 2024. Available at: https://mc-stan.org/cmdstanr/. Accessed March 27, 2026.

[26] Alsanie I, Rajab S, Cottom H, Adegun O, Agarwal R, Jay A, et al. Distribution and frequency of salivary Gland tumours: an international multicenter Study. Head Neck Pathol 2022;16(4):1043.

[27] Vermeersch P, Bossuyt X. Comparative analysis of different approaches to report diagnostic accuracy. Arch Intern Med 2010;170:734−5.

[28] Stoye FV, Tschammler C, Kuss O, Hoyer A. A discrete time-to-event model for the meta-analysis of full ROC curves. Res Synth Methods 2024;15(6):1031−48.

[29] Rücker G, Steinhauser S, Kolampally S, Schwarzer G. Diagmeta: Meta-Analysis of diagnostic accuracy studies with several cutpoints. R Package. 2022. Available at: https://cran.r-project.org/package=diagmeta. Accessed March 27, 2026.

[30] Hoyer A, Hirt S, Kuss O. Meta-analysis of full ROC curves using bivariate time-to-event models for interval-censored data. Res Synth Methods 2018;9(1):62−72.

[31] Hamza TH, Arends LR, Van Houwelingen HC, Stijnen T. Multivariate random effects meta-analysis of diagnostic tests with multiple thresholds. BMC Med Res Methodol 2009;9:1−15.

[32] Putter H, Fiocco M, Stijnen T. Meta-analysis of diagnostic test accuracy studies with multiple thresholds using survival methods. Biometrical J 2010;52(1):95−110.

[33] Zwinderman AH, Bossuyt PM. We should not pool diagnostic likelihood ratios in systematic reviews. Stat Med 2008;27:687−97.

[34] Leeflang MMG, Deeks JJ, Gatsonis C, Bossuyt PMM. Systematic reviews of diagnostic Test accuracy. Ann Intern Med 2008;149: 889.

[35] Talts S, Betancourt M, Simpson D, Vehtari A, Gelman A. Validating bayesian inference algorithms with simulation-based calibration. arXiv. 2020. Available at: https://arxiv.org/abs/1804.06788. Accessed March 27, 2026.